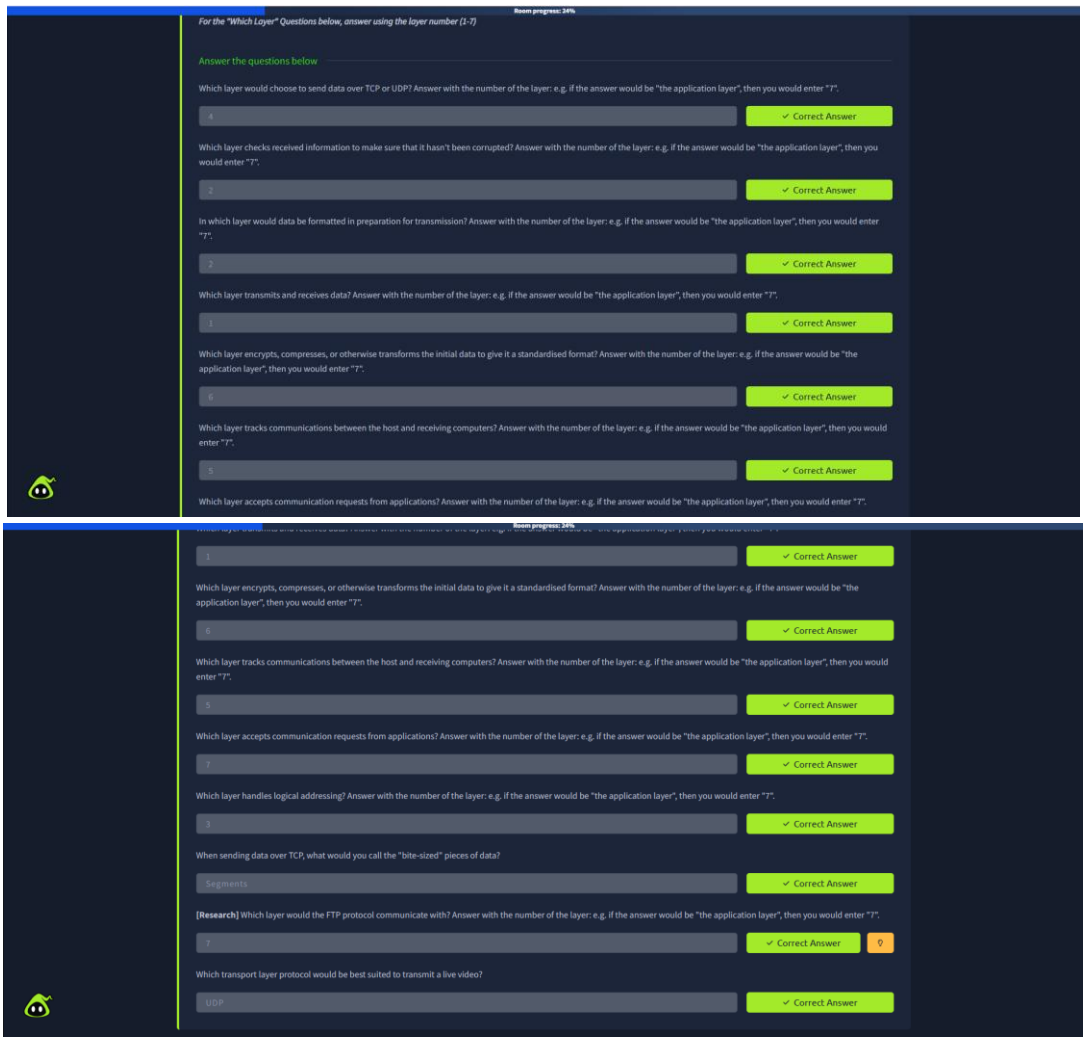
TryHackMe Writeup: Introductory Networking

Room: OSI and TCP/IP models, encapsulation, TCP handshake, ping, traceroute, WHOIS, DNS and dig

Objective: Learn the OSI model, TCP/IP model, encapsulation, and basic networking tools.

Step 1 - Memorize the seven OSI layers
The OSI model is a conceptual framework for understanding network communication.

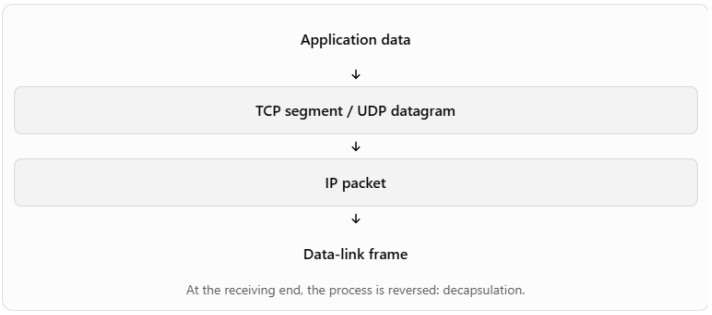
7. Application - Network services used by applications
6. Presentation - Data representation, encryption, compression
5. Session - Managing communication sessions
4. Transport - End-to-end transport, ports, TCP and UDP
3. Network - IP addressing and routing
2. Data Link - Local delivery and frames
1. Physical - Signals and physical transmission

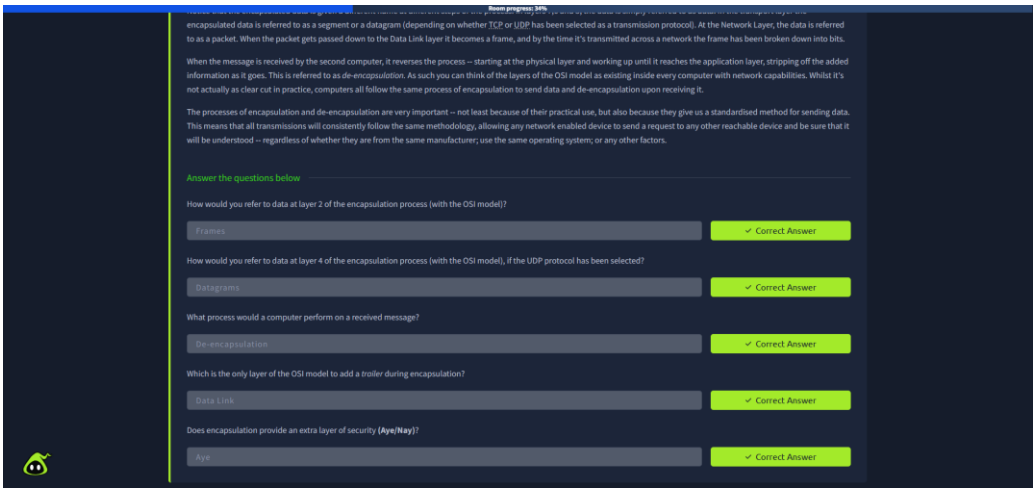


Step 2 - Learn data encapsulation

When data is sent, each networking layer adds the information required for its part of the journey.

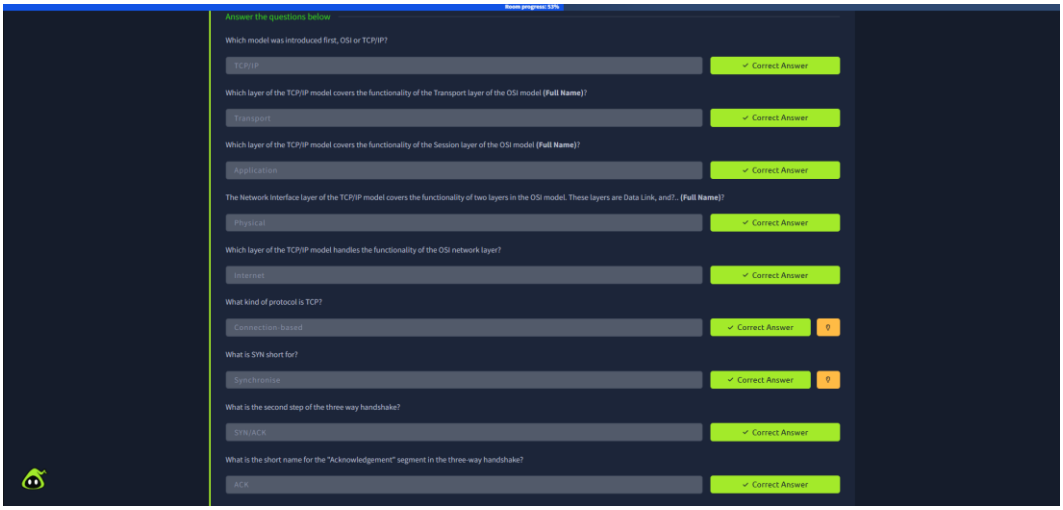
Layer / stage	Name of data
Application, Presentation, Session	Data
Transport with TCP	Segment
Transport with UDP	Datagram
Network	Packet
Data Link	Frame
Physical	Bits





Step 3 - The TCP/IP Model

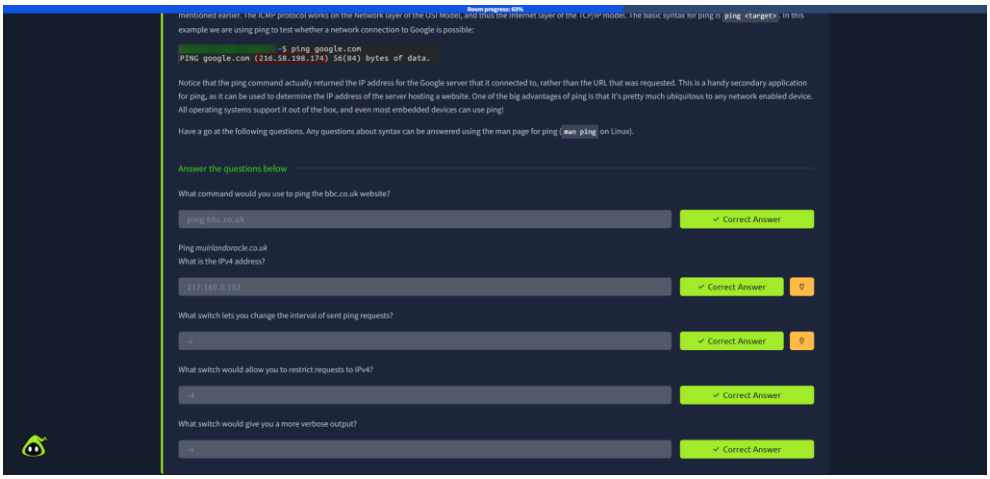
The common four-layer TCP/IP model consists of Application, Transport, Internet and Network Interface. It groups some functions differently from the seven-layer OSI model.



The TCP three-way handshake is **SYN → SYN/ACK → ACK**. TCP establishes a connection and supports reliable, ordered delivery.

Step 4 - Networking Tool: Ping

ping sends probe messages to check whether a destination responds and to estimate round-trip time. It commonly uses ICMP.



Practical commands

ping bbc.co.uk

ping muirlandoracle.co.uk

man ping

The DNS address and ping result can vary over time. A host may also block ICMP, so no response does not always mean the host is offline.

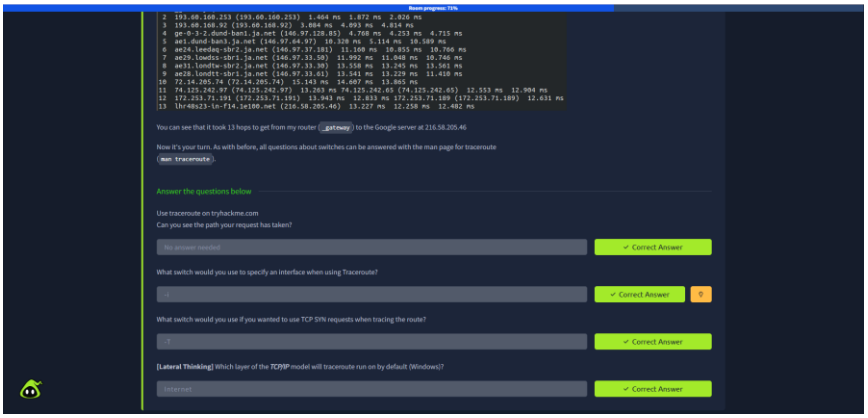
```
C:\Users\Prabhakar>Ping muirlandoracle.co.uk

Pinging muirlandoracle.co.uk [2001:8d8:100f:f000::241] with 32 bytes of data:
Reply from 2001:8d8:100f:f000::241: time=177ms
Reply from 2001:8d8:100f:f000::241: time=167ms
Reply from 2001:8d8:100f:f000::241: time=164ms
Reply from 2001:8d8:100f:f000::241: time=164ms

Ping statistics for 2001:8d8:100f:f000::241:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 164ms, Maximum = 177ms, Average = 168ms
```

Step 5 - Networking Tool: Traceroute

traceroute helps show the route packets take toward a destination by eliciting responses from intermediate network hops. Some hops may not respond, so * entries are possible.



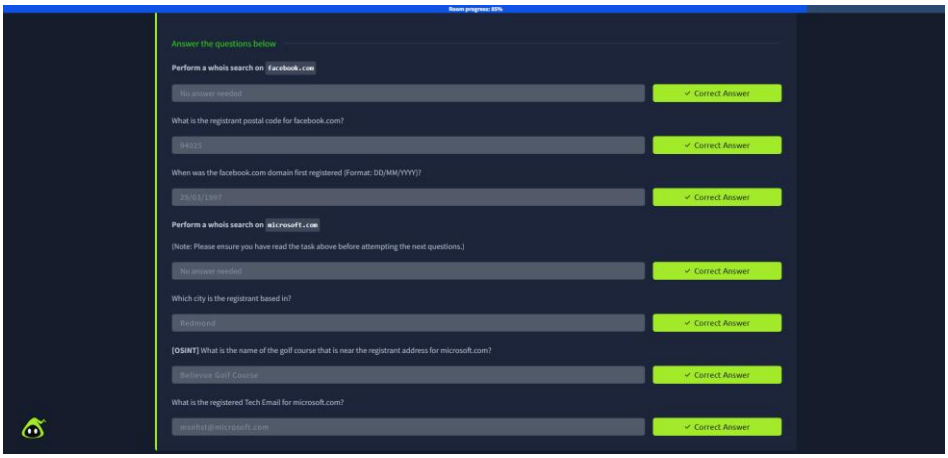
Step 6 - Networking Tool: WHOIS

WHOIS provides registration-related information about domain names, such as registrar data, registration dates, name servers, and registrant details when publicly available.

whois facebook.com

whois microsoft.com

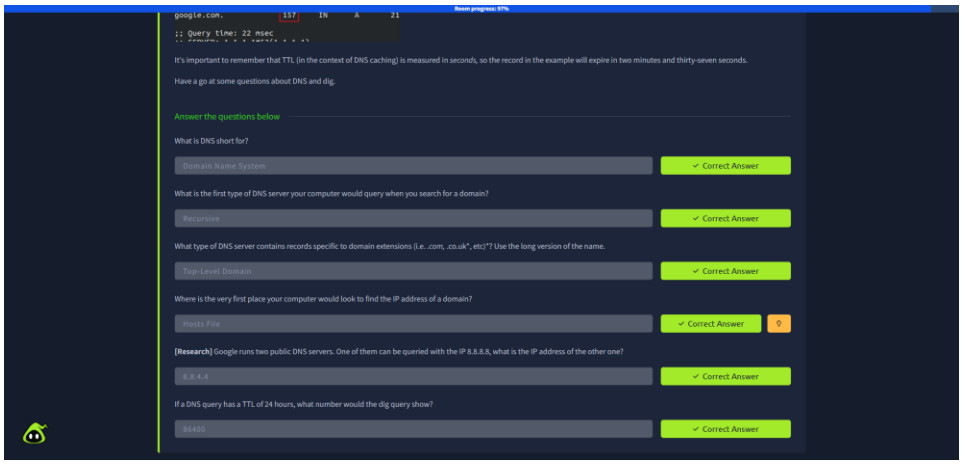
If whois is not installed in the Linux environment, use the web WHOIS tool provided by the room or an approved public lookup service.



WHOIS information can change or be redacted by registrars. Use the output and instructions in your live room if the values differ.

Step 7 - Networking Tool: dig and DNS

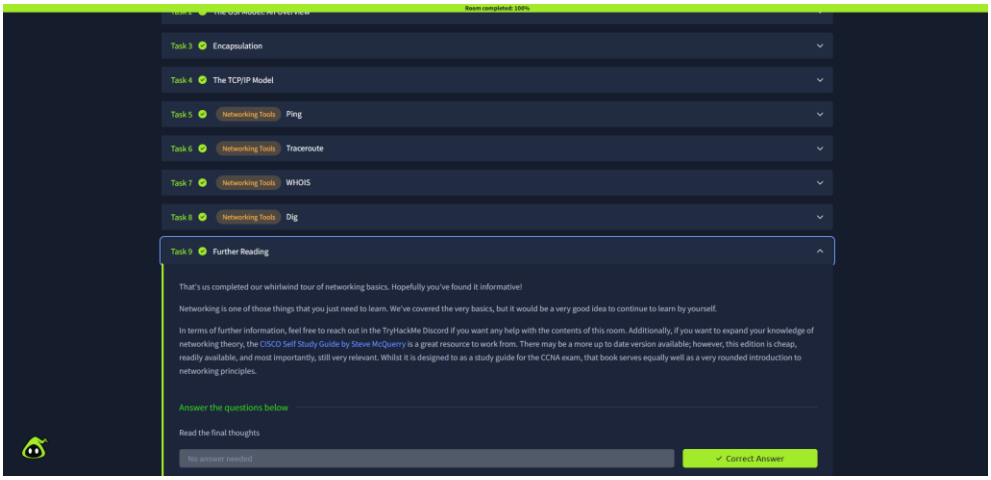
DNS (Domain Name System) maps domain names to resource records such as IP addresses. A resolver usually checks applicable local sources and caches before querying other DNS servers.



Note on lookup order: The room's explanation describes checking the hosts file before the DNS cache. Some walkthroughs phrase this answer as “local cache”; use the wording that matches the live room's current expected answer.

Step 8 - Further Reading

Read the room's final thoughts and mark the task complete in TryHackMe. No separate technical answer is required; follow the room's completion instructions.



What you should learn: The OSI model explains the stages of communication, while the TCP/IP model describes the protocol architecture commonly used on real networks.